



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RAJ FUEL SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

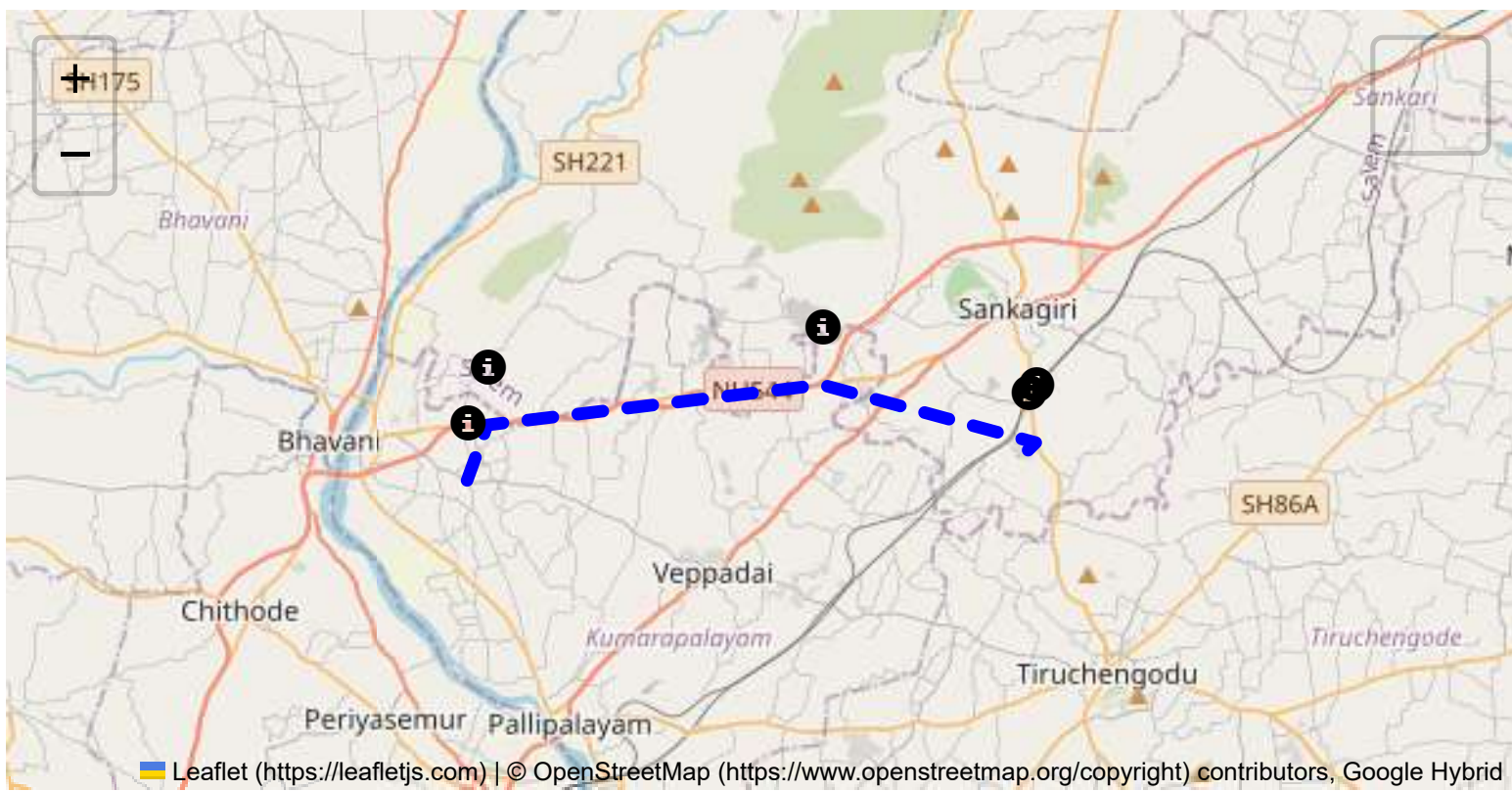
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

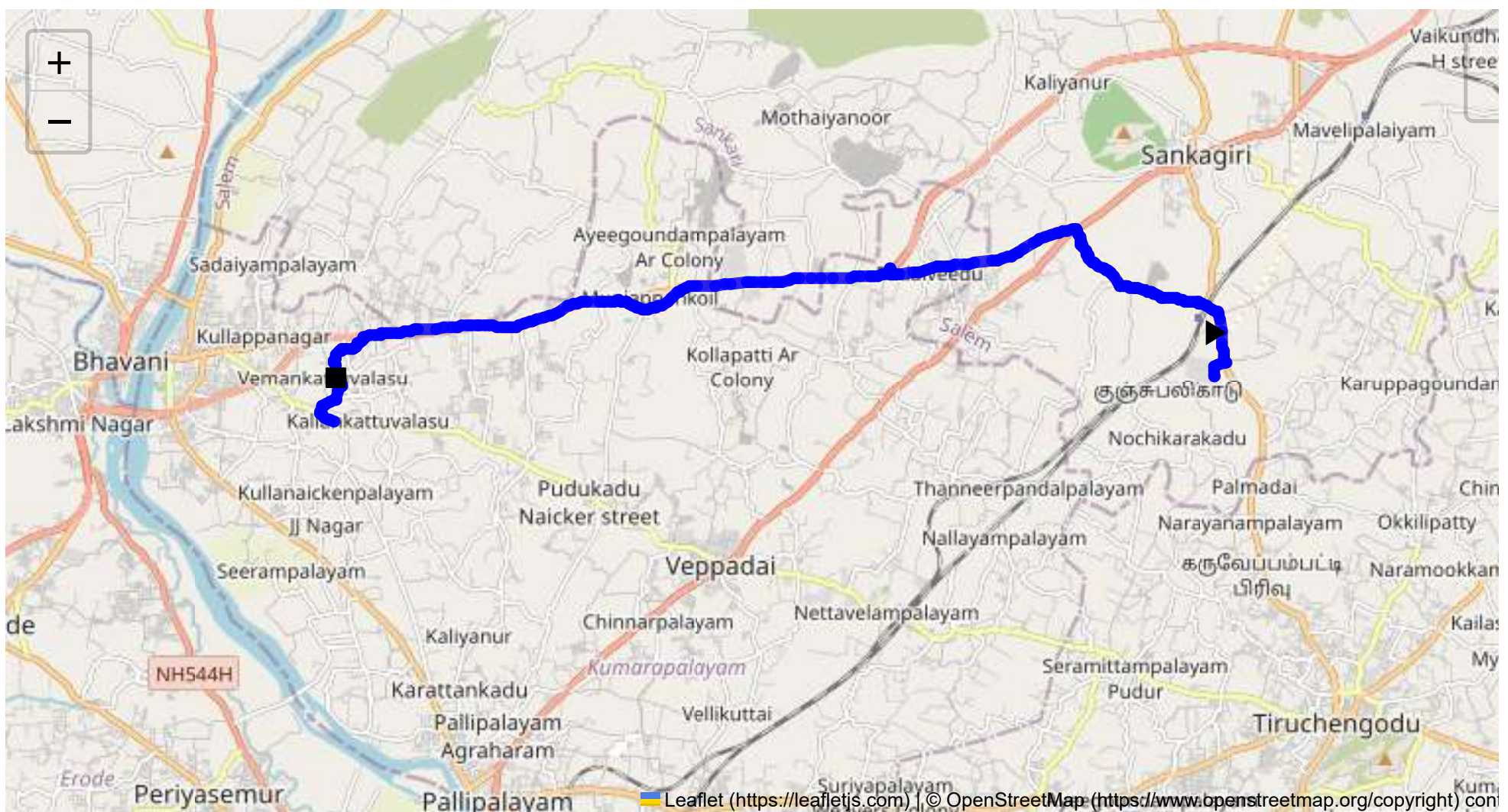
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 21.44 km
Estimated Duration: 0.5 hours
Adjusted Duration (Heavy Vehicle): 0.6 hours
Start: (11.4381, 77.8734)
End: (11.43033, 77.71983)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Pallipalayam spans approximately 21.44 kilometers, traveling through several villages and towns, including Kangeyampalayam, Padaiveedu, and along Service Road in Tamil Nadu. This route primarily consists of state highways and connecting roads that facilitate inter-town travel.

Typical Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate with hot summers, moderate monsoons, and cool winters. The route can be affected by monsoonal rains (June to September), leading to slippery roads and possible waterlogging. Fog is uncommon but may occur during winter mornings, reducing visibility.

Traffic Patterns

Traffic is typically moderate on this route with potential congestion:

- **Morning Peak Hours:** 8:00 AM - 10:00 AM
- **Evening Peak Hours:** 5:00 PM - 7:00 PM

Congestion-prone areas include town centers and busy intersections, particularly in Sangagiri and Pallipalayam.

Road Quality and Infrastructure

Road quality along this route varies:

- **Sangagiri to Kangeyampalayam:** Generally well-maintained with clear signage.
- **Kangeyampalayam to Padaiveedu:** Narrower roads and potential for potholes, especially post-monsoon.
- **Padaiveedu to Pallipalayam:** Mixed quality with some repair work necessary.

Alternative Routes for Emergencies

In cases of roadblocks or major incidents, consider the following alternatives:

- **Via SH-86:** Connects through nearby towns, slightly longer but avoids smaller roads.
- **Via NH-544:** Major highway route offering better road conditions and faster response times.

Local Regulations on Hazardous Materials

Transport of hazardous materials in Tamil Nadu is regulated by the state transport department:

- Ensure adherence to safety protocols (e.g., clear labeling and safety gear).
- Restrictions may apply during night travel in some areas due to safety concerns.

Historical Incidents

There are limited records of significant incidents along the specified route involving heavy vehicles or hazardous materials. However, sporadic issues like overturned trucks or minor spills have been reported, generally due to driver error or road quality issues.

Environmental Considerations

The route does traverse through semi-rural areas with occasional proximity to agricultural lands. Ensure no spills occur, as they might impact local livelihood and environment.

Communication Coverage

Mobile network coverage is generally good, but occasional dead zones may exist, particularly in less populated areas or stretches with geographical barriers.

Estimated Emergency Response Times

- Urban Areas (Sangagiri, Pallipalayam): 15-30 minutes
- Rural Stretches (Kangeyampalayam, Padaiveedu): 30-60 minutes

Emergency services like hospitals and fire stations are accessible, but response times can vary based on location and time of day.

Overall Risk Assessment Summary

While the route presents moderate risk for transporting hazardous materials, proper planning and adherence to safety standards can mitigate most issues. Awareness of peak times, alternative routes, and environmental considerations is crucial for safe passage. Maintaining clear communication, especially in potential dead zones, and understanding local regulations will ensure a safer journey. Emergency response times are generally reasonable, but drivers should always be prepared for delayed assistance in rural sections. Proper training and adherence to procedures are essential for minimizing risk.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45348, 77.85706	30 KM/Hr
6	Turn	Medium	11.46318, 77.84913	30 KM/Hr
7	Turn	Medium	11.45697, 77.82719	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
10	Turn	High	11.45636, 77.81678	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Blind Spot	Blind Spot	11.45543, 77.81740	10 KM/Hr
12	Turn	High	11.45516, 77.81346	15 KM/Hr
13	Turn	High	11.45521, 77.81343	15 KM/Hr
14	Blind Spot	Blind Spot	11.44530, 77.72783	10 KM/Hr
15	Blind Spot	Blind Spot	11.44513, 77.72786	10 KM/Hr
16	Turn	High	11.44474, 77.72485	15 KM/Hr
17	Turn	High	11.43618, 77.72111	15 KM/Hr
18	Turn	High	11.43408, 77.72009	15 KM/Hr
19	Turn	High	11.43167, 77.71713	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital	11.4525596, 77.7749426	30 km/h	Medium
4	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium
5	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
6	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45348, 77.85706



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45697, 77.82719



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45516, 77.81346



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45521, 77.81343



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44530, 77.72783



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44513, 77.72786



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44474, 77.72485



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43618, 77.72111



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43408, 77.72009
